
Business Case

**Reducing Turnover Through Workforce
Planning & People Analytics**

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CASE

ALPHA MANUFACTURING

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1. Business Context

Company: Alpha Manufacturing (fictional company)

Industry: Automotive Manufacturing

Employees: 1.200

Challenge:

During the first quarter of 2025, the company observed an increase in voluntary turnover, particularly among production employees. Leadership requested a People Analytics study to identify the main drivers and recommend actions to reduce turnover while maintaining

2. Objectives

- Analyze workforce demographics.
- Identify turnover patterns.
- Understand the relationship between onboarding, tenure and turnover.
- Evaluate absenteeism.
- Provide data-driven recommendations.

3. Data Overview

- **Dataset:**
 - 1.200 employees
 - HRIS data
 - Attendance records
 - Organizational structure
 - Workforce Planning indicators
- **KPIs:**
 - Headcount
 - Turnover Rate
 - Occupancy Rate
 - Absenteeism Rate
 - Gender Distribution
 - Age Distribution
 - Distance from Home
 - Tenure
 - Department
 - Shift

4. Executive Summary

The analysis of Alpha Manufacturing's workforce identified important patterns related to turnover, workforce distribution, and absenteeism. While the overall workforce remains balanced across departments and demographic groups, employee retention during the first year of employment emerged as the company's main challenge.

Turnover was significantly higher among employees with less than one year of service, particularly in Shift 2 and in production departments. Employees living farther from the facility also showed higher turnover rates, suggesting that commuting distance may influence retention.

The analysis indicates a strong relationship between onboarding completion and employee retention. Departments with lower onboarding completion rates experienced higher turnover, reinforcing the importance of an effective integration process during the first months of employment.

Absenteeism remained concentrated in specific operational areas, indicating opportunities to investigate workload distribution, attendance management, and local leadership practices.

Based on these findings, the recommended priorities are to strengthen onboarding, improve first-year retention initiatives, monitor critical workforce indicators through People Analytics dashboards, and support managers with data-driven insights for workforce planning decisions.

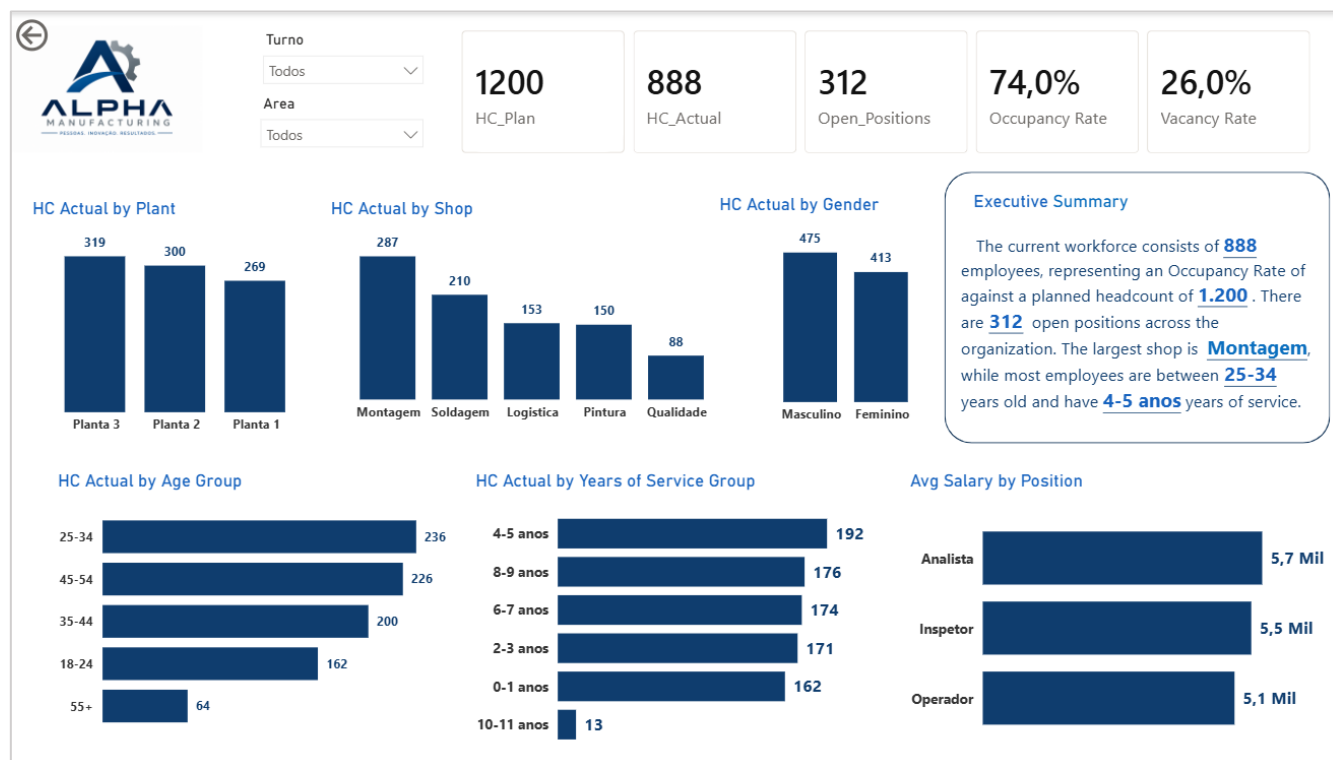
These actions are expected to improve workforce stability, reduce replacement costs, and enable more proactive and strategic talent management.

5. Dashboard Overview

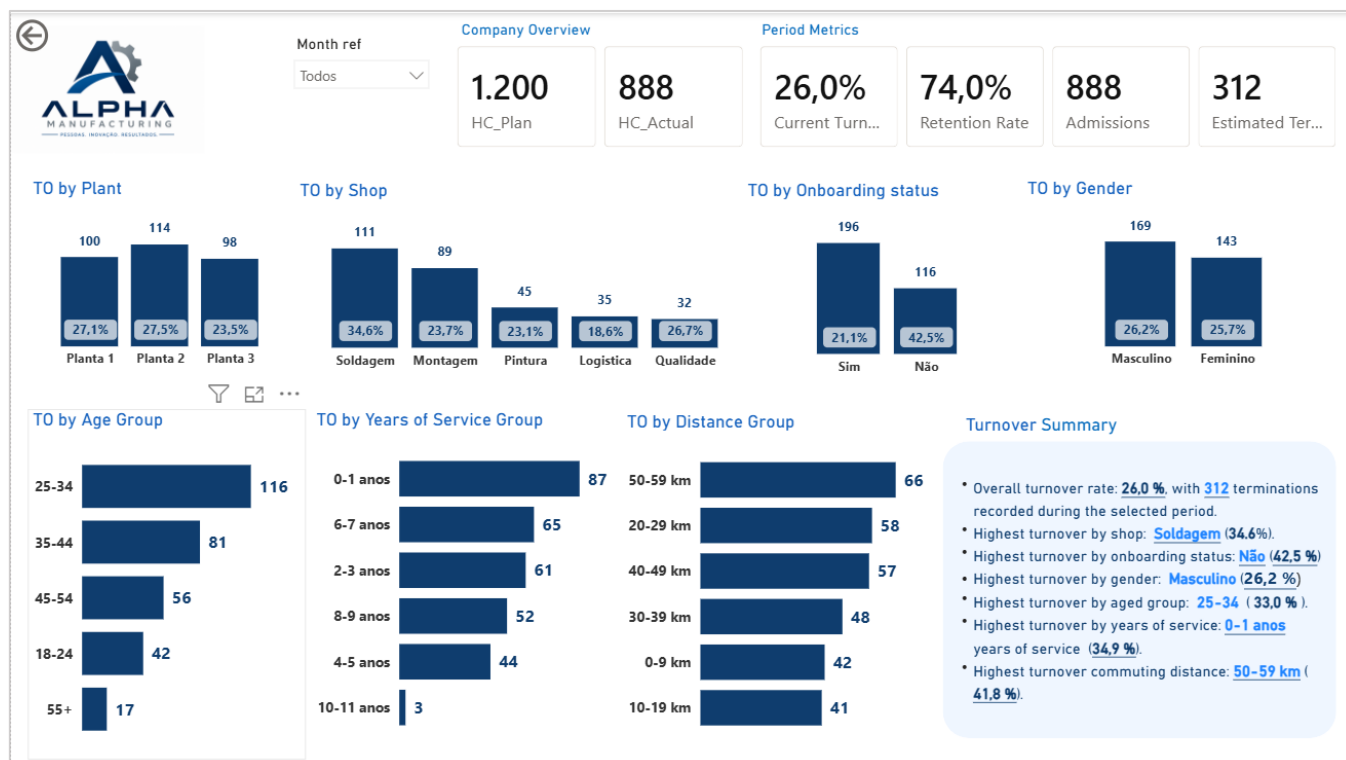
Power BI Dashboard



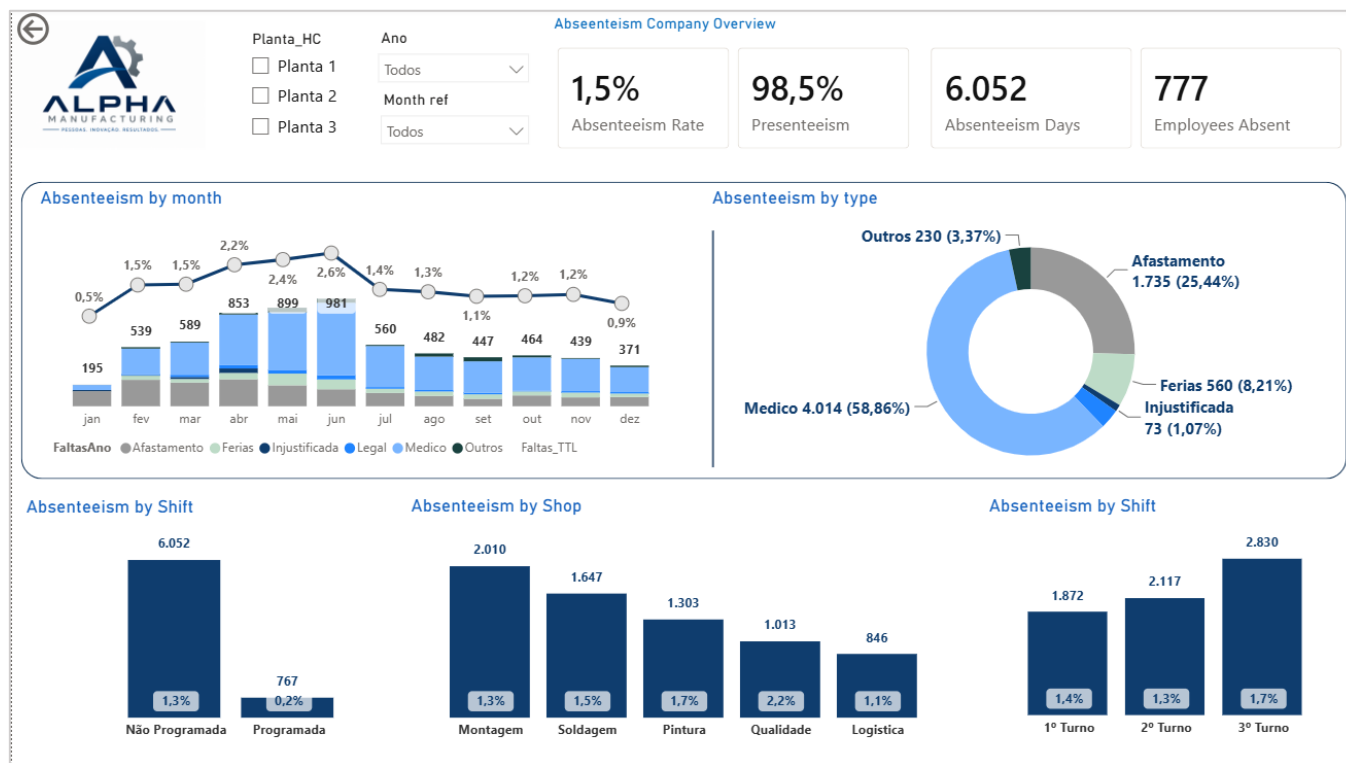
Workforce Overview



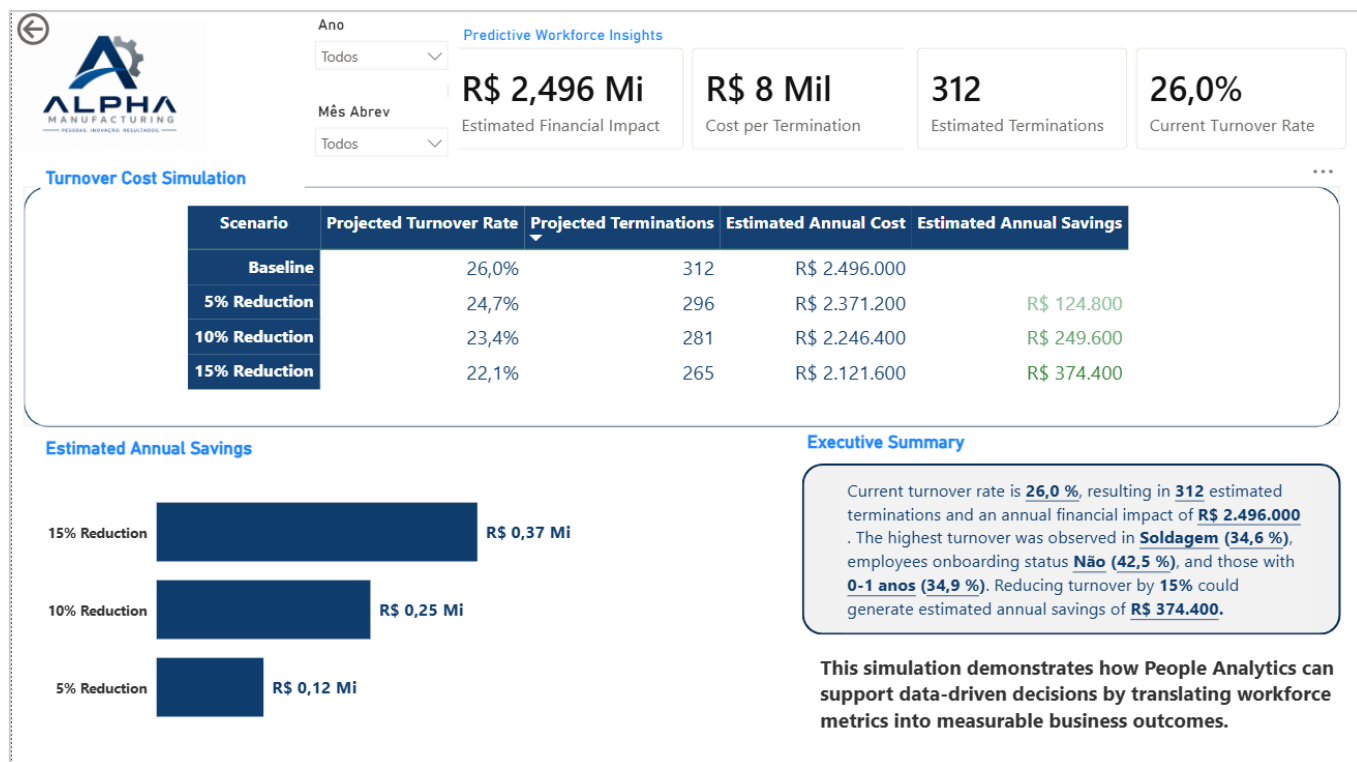
Turnover Analytics



Absenteeism Analytics



Predictive Analytics



6. Key Findings

Turnover

- Employees with less than one year of tenure showed the highest turnover.
- Shift 2 presented the highest turnover rate.
- Employees living farther from the plant showed greater turnover.
- Departments with lower onboarding completion also experienced higher turnover.

Absenteeism

- Higher absenteeism was concentrated in specific production areas.
- Shift patterns suggest operational and scheduling impacts.
- Longer commuting distances may contribute to absenteeism.

7. Recommendations

- **Short Term**

- Strengthening onboarding.
- Review first-year retention practices.
- Monitor Shift 2 weekly.
- Implement monthly turnover reviews.

- **Medium Term**

- Improve transportation support.
- Enhance supervisor training.
- Develop retention programs for critical positions.

- **Long Term**

- Predict turnover using People Analytics.
- Build workforce planning simulations.
- Create executive HR dashboards.

8. Business Impact

- **Potential benefits include:**

- Reduced turnover costs.
- Improved workforce stability.
- Better productivity.
- Faster decision-making.
- Increased employee retention.

9. Technical Skills Demonstrated

- Power BI
- Power Query
- DAX
- Excel
- HR Analytics
- Workforce Planning
- Dashboard Design
- Data Storytelling
- KPI Development

10. Disclaimer

Disclaimer:

This project was developed exclusively for portfolio purposes. The company, data, and scenarios are fictional but were designed to simulate real business situations using People Analytics and Workforce Planning methodologies.

About the Author



Priscila Lima Gomes

Priscila Lima has more than 15 years of experience in Workforce Planning and People Analytics across complex manufacturing environments.

She specializes in transforming workforce data into strategic insights that support business decisions through dashboards, KPI development, workforce planning, and data-driven storytelling.